

Khaled Hasan

+880-1622-156330

khaledirfan7@gmail.com

[Khaled Hasan](#)

[GitHub](#)

[Khaled Hasan](#)

[Website](#)

Mirpur, Dhaka-1216, Bangladesh

SUMMARY

Computer Science lecturer and researcher developing human-centered, trustworthy AI systems that pair machine learning and large language/visionon-language models, explainable AI with rigorous human-centered evaluation, so that real-world decision-support and interactive systems become more reliable, interpretable, and usable across adaptive user interfaces, AI for health and edge intelligence.

Research Interests: Trustworthy & Explainable AI; Human-AI Interaction and Human-Centered ML; AI Systems; Large Language & Vision-Language Models; Applied Machine Learning for Health; Edge & On-Device Intelligence.

EDUCATION

• Military Institute of Science and Technology

Jan 2020 - Mar 2024

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

◦ CGPA: 3.83 out of 4.00 (Highest GPA: 3.98 out of 4.00, Spring 2022)

◦ 4th out of 88 students in the whole department

◦ **Thesis Title: Adaptive User Interface in Mobile Banking: A Machine Learning Approach to Enhance UX**

Developed an adaptive UI prototype using a customized ML pipeline based on users' tapping behavior for enhancing the UX of the mobile apps used by the banking organizations and their users.

• Notre Dame College

Jul 2017 - Apr 2019

Higher Secondary Certificate

Dhaka, Bangladesh

◦ GPA: 5.00 out of 5.00 (Board Scholarship)

• Ideal School and College

Jan 2007 - Feb 2017

Secondary School Certificate

Dhaka, Bangladesh

◦ GPA: 5.00

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, T=THESIS

Journal Articles (Published)

[J1] **Khaled Hasan**, Md Rashid Ul Islam, *et al.* (2026). "Adaptive User Interface for Mobile Banking Apps: Enhancing UX through Machine Learning." *Array* (Elsevier), 2026, Art. 100901. [Q1, IF: 4.5]. [doi:10.1016/j.array.2026.100901](https://doi.org/10.1016/j.array.2026.100901)

Conference Papers (Published)

[C1] **Khaled Hasan**, Maliha Zerin, *et al.* (2024). "Light Canvas: An IoT-Based Interactive Virtual Drawing System for Enhancing Educational Engagement." *Proceedings of the 3rd International Conference on Computing Advancements (ICCA 2024)* (ACM). [doi:10.1145/3723178.3723192](https://doi.org/10.1145/3723178.3723192)

[C2] **Khaled Hasan**, Adib Hossain, *et al.* (2025). "From Keywords to Context: A Comparative Analysis of Semantic (SBERT) vs Traditional Keyword-Based (Machine Learning) Matching for Job Recommendation." *28th International Conference on Computer and Information Technology (ICCIT)*, IEEE. [doi:10.1109/ICCIT68739.2025.11491820](https://doi.org/10.1109/ICCIT68739.2025.11491820)

[C3] Md Rashid Ul Islam, **Khaled Hasan**, *et al.* (2025). "Assessing Usability of Agricultural Mobile Applications for Illiterate and Semi-Literate Farmers in Bangladesh." *28th International Conference on Computer and Information Technology (ICCIT)*, IEEE. [doi:10.1109/ICCIT68739.2025.11491675](https://doi.org/10.1109/ICCIT68739.2025.11491675) - **Top 2% Paper Award in the conference.**

Conference Papers (Accepted / In Press)

[C4] **Khaled Hasan**, Alisha Kabir, *et al.* (2025). "Evaluating User Experience in Decentralized Social Media Platforms: A Comparative Study of Web3 and Web2 Social Media Apps." *HCI-E 2025*. Accepted and presented (to appear in Springer CCIS proceedings).

Manuscripts (Under Review)

[J2] Md Rosaidul Mawla, **Khaled Hasan**, *et al.* (2025). "Protective Zoning and Predictive Modeling for Nuclear Safety: A Machine Learning Approach at Rooppur." Under review, *Journal of Nuclear Materials* (Q1).

ONGOING RESEARCH

- **Trustworthy & Explainable Skin-Lesion Classification:** Extending a hybrid DINO-ViT (self-supervised vision transformer) and quantum-annealing feature/hyperparameter-optimization pipeline on HAM10000 toward a clinician-in-the-loop tool adding uncertainty quantification, visual explanations, and fairness across skin tones for reliable computer-aided dermatological diagnosis.
- **Human-Centered Evaluation & Reliability of Vision-Language Models:** Studying when and why users over- or under-trust multimodal LLM outputs, and designing interface and calibration strategies that make vision-language models dependable for low-resource and accessibility-focused decision support.
- **Air Quality & Airborne-Disease Burden Prediction:** Modeling the correlation between air pollution and airborne diseases to forecast disease burden using machine learning over sensor and clinical data.
- **Technical vs Non-Technical User Interaction with various LLMs:** A comparative, human-centered study of how user expertise shapes interaction patterns, prompting behavior, and outcomes across conversational LLM assistants.

PROFESSIONAL EXPERIENCE

- **Lecturer, Dept. of Computer Science and Engineering**
Military Institute of Science and Technology, Dhaka, Bangladesh Apr 2024 – Present
 - Conduct applied-ML and HCI research alongside teaching; mentor undergraduate research and capstone-thesis projects, several leading to peer-reviewed publications.
 - Guided 220+ students across theory and laboratory courses in programming, data structures, and ML application development.
- **Full-Stack Developer, MIST CSE Archive**
Military Institute of Science and Technology, Dhaka, Bangladesh May 2022 – Dec 2023
 - Developed and maintained an academic archive web portal used by over 800 students.
- **Industrial Trainee, IT Department**
Grameenphone Ltd., Dhaka, Bangladesh Jan 2023 – Feb 2023
 - Assisted in building dashboards and backend automations.
 - Collaborated with teams to streamline operational data pipelines.

TEACHING EXPERIENCE

Programming & Software: CSE-105 Structured Programming Language, CSE-106 Structured Programming Language Sessional, CSE-110 Computer Programming Laboratory, CSE-205 Object-Oriented Programming Language, CSE-206 OOP Sessional-I, CSE-220 OOP Sessional-II, CSE-216 Data Structures & Algorithms Sessional-II, CSE-281 Computer Programming, CSE-282 Computer Programming Sessional, CSE-291 Computer Programming Techniques, CSE-292 Computer Programming Techniques Sessional.

Systems & Architecture: CSE-104 Digital Logic Design Laboratory, CSE-213 Computer Architecture, CSE-307 Operating System, CSE-308 Operating System Sessional, CSE-310 Computer Networking Sessional.

Capstone Supervision: CSE-350 Industrial Training, CSE-400 Undergraduate Thesis.

SKILLS

- **Programming:** Python, C++, C, JavaScript, SQL, Bash, LaTeX
- **ML / AI:** PyTorch, TensorFlow, Keras, Scikit-learn, FastAI, HuggingFace Transformers, OpenCV, NLTK, spaCy
- **Deployment & Tools:** FastAPI, Streamlit, Git, Raspberry Pi, Arduino IDE, Google Colab, Jupyter Notebook, Figma, Tableau, MATLAB, Cisco Packet Tracer
- **Database Systems:** MySQL, MongoDB, Firebase, Oracle

NOTABLE PROJECTS

- **Light Canvas: A Ubiquitous Interactive Virtual Drawing System** Jul 2022 - Jun 2023
Technology: Image Processing, Machine Learning, Figma, Flutter, Android Studio, Dart, Python, OpenCV, IoT.  
 - Developed an IoT-based portable device that turns any flat surface into a virtual drawing board by projecting a laser surface, aiding children's education without complex computational devices.
- **AutoML Tool for Airborne Disease Analysis** Jan 2023 - Jun 2023
Technology: Python, Arduino, Pressure Sensors 
 - Developed an AutoML tool analyzing the correlation between air pollution and airborne diseases, predicting disease burden.
- **NLP-Based AI Job Recommendation System** Jul 2023 - Dec 2023
Technology: NLP, Machine Learning, Python 
 - Developed an NLP-based job recommendation system where candidates upload their CVs and job preferences to receive AI-driven job recommendations.
- **Evaluating User Experience in Decentralized Social Media Platforms: Web3 vs Web2** Jul 2023 - Feb 2024
Technology: HCI, SUS 

- Designed a prototype to evaluate UX differences between Web3 and Web2 social media platforms using key UX parameters for improvement.
- **Line Follower Delivery Bot** Jul 2022 - Dec 2022
Technology: Arduino, IoT Sensors 
◦ Developed a prototype delivery bot with a chamber carrying up to 3kg, which followed a predefined path using line-following technology.

LEADERSHIP & SERVICE

- **Project Coordinator, Mobile & App Dev. Project for Bangladesh Army** May 2025 – Present
Military Institute of Science and Technology, Dhaka, Bangladesh
◦ Overseeing full-cycle design and deployment of a secure mobile app suite for internal Army use.
◦ Coordinating a team of developers, UI/UX designers and testers.
- **Moderator, MIST Cyber Security Club** Jul 2024 – Present
Military Institute of Science and Technology, Dhaka, Bangladesh
◦ Organizing national-level cybersecurity workshops and internal CTFs; managing all technical operations of the club.
- **Executive Director, MIST Computer Club** Feb 2023 – Mar 2024
Military Institute of Science and Technology, Dhaka, Bangladesh
◦ Organized student-led tech initiatives including coding competitions and seminars.
- **Software & Communication Team, MIST Mars Rover Society** Mar 2020 – Oct 2023
Military Institute of Science and Technology, Dhaka, Bangladesh
◦ Developed embedded autonomous control software for the rover Phoenix 2.0.
◦ Participated in URC and ARC international robotics competitions.

HONORS AND AWARDS

- **Top 2% Paper Award ICCIT 2025 (IEEE)** Dec 2025
28th International Conference on Computer and Information Technology
[Certificate](#)
◦ For the paper “Assessing Usability of Agricultural Mobile Applications for Illiterate and Semi-Literate Farmers in Bangladesh.” Recognized among the top 2% of all accepted papers.
- **Commandant's List of Honour** Jan 2020 – Apr 2024
Military Institute of Science and Technology
[Link](#)
◦ Achieved CGPA 3.80 or above in 2020, 2022, and 2023.
- **Dean's List of Honour** Jan 2020 – Apr 2024
Military Institute of Science and Technology
[Certificate](#)
◦ Achieved CGPA 3.70 or above in all four academic years.
- **Project Demo Selected for Digital Display at National Innovation Summit** Jul 2025
BEAR Summit 2025, Bangladesh National Semiconductor Symposium
[Certificate](#)
◦ Team Lead of the project.
- **Champion at Research-Based Poster Presentation** Aug 2023
BUP IEEE Branch Fest
[Certificate](#)
◦ Team Lead of Team Light Canvas.
- **Champion University Rover Challenge 2021** Jul 2021
The Mars Society, USA
[Certificate](#)
◦ Team Member of MIST MONGOL BAROTA.
- **2nd Runner-up Anatolian Rover Challenge 2022** Jul 2022
Space Exploration Society (UKET), Türkiye
[Certificate](#)
◦ Team Member of MIST MONGOL BAROTA.
- **Top 10 Teams Inter-University Hackathon** Jul 2023
BUET CSE Fest
[Link](#)
◦ Team member of Team MISTCarDNulls.
- **Top 20 Teams Final Round WIEHACK 4.0 Hackathon** Jul 2023
Women in Engineering, IEEE Bangladesh
[Certificate](#)
◦ Team Lead of Team Light Canvas.

REFERENCES

1. **Dr. Md. Mahbubur Rahman**
Professor, Computer Science and Engineering
Military Institute of Science and Technology
Email: mahbub@cse.mist.ac.bd
2. **Dr. Muhammad Nazrul Islam**
Former Associate Professor, Computer Science and Engineering
Military Institute of Science and Technology
Email: nazrul@cse.mist.ac.bd
3. **Dr. T. M. Shahriar Sazzad**
Associate Professor, Computer Science and Engineering
Military Institute of Science and Technology
Email: shahriar@cse.mist.ac.bd